Unit 3

Electronic Payment System

- **E-payment System:** Online Credit Card Transaction, Online Stored Value Payment System, Digital and Mobile Wallet, Smart Cards, Social/Mobile Peer-to-Peer Payment Systems, Digital Cash/e-cash, E-Checks, Virtual Currency
- Electronic Billing Presentment and Payment (EBPP) System, Auctioning in E-commerce (English, Dutch, Vickery, Double),
- SET Protocol, Features of SET, Participants in SET, Card Holder Registration, Merchant Registration, Purchase Request, Dual Signature, Payment Authorization, Payment Capture
- Status of E-Payment Systems in Nepal

Link for unit 1&2 slides :

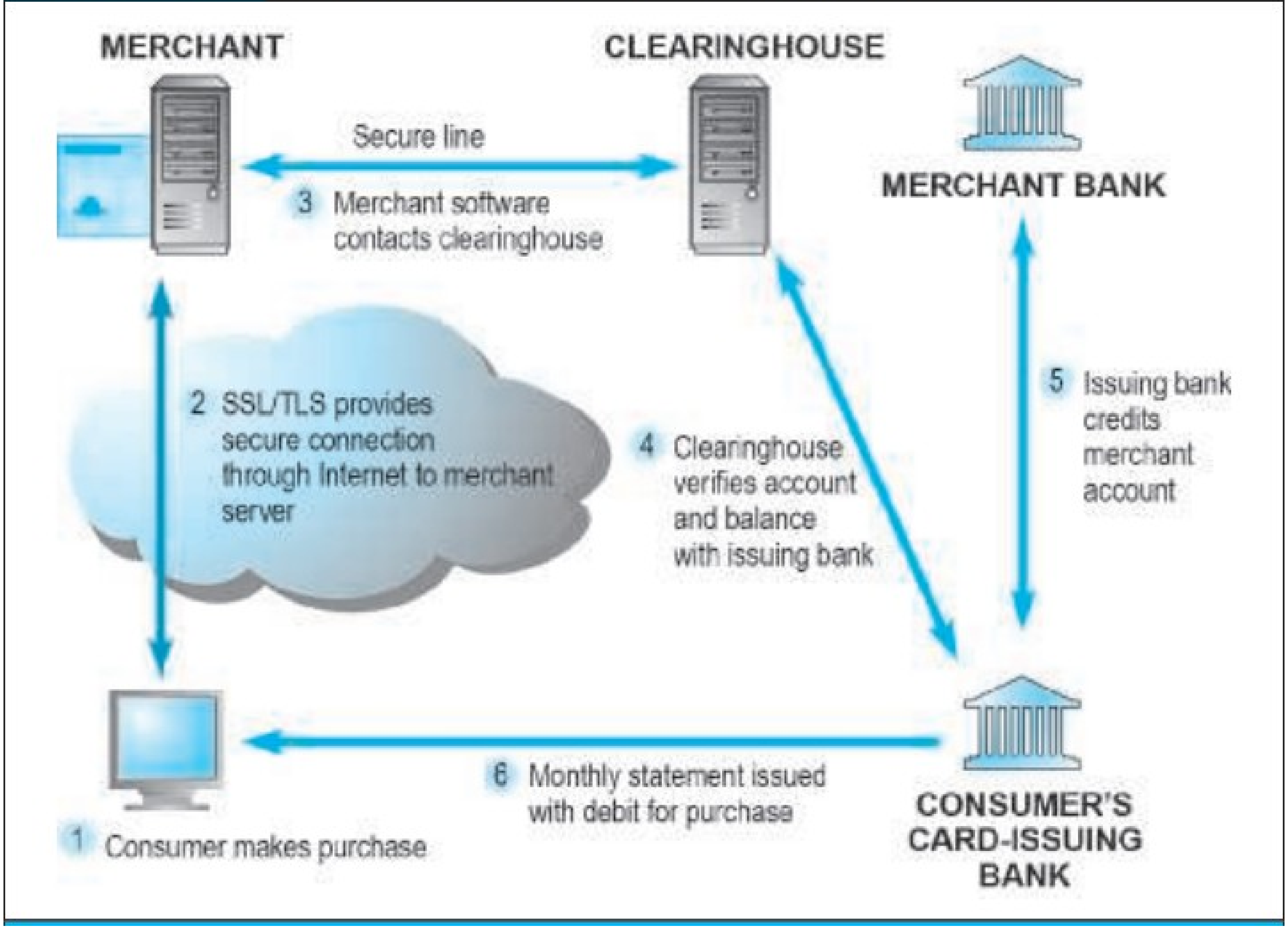
https://drive.google.com/file/d/15KcwJk-fp_myxgUVPVMSi2Awe2w9fyXT/view?usp=sharing

- In any commercial transaction(buying/selling) , payment is an integral and most important part (along with order and delivery).
- In traditional commerce, the most widely used payment method is paper cash or paper (bank) cheque.
- These traditional methods of payment are not suitable for ecommerce because we can not pay online using paper cash.
- payment mechanisms such as cash, credit cards, debit cards, checking accounts have been able to be adapted to the online environment.
- New types of purchasing relationships, such as between individuals online, and new technologies, such as the development of the mobile platform, have also created both a need and an opportunity for the development of new payment systems.

ONLINE CREDIT CARD TRANSACTIONS

- Because credit and debit cards are the dominant form of online payment
- Online credit card transactions are processed in much the same way that in-store purchases are, with the major differences being that online merchants never see the actual card being used, no card impression is taken, and no signature is available.
- These types of purchases are also called Cardholder Not Present (CNP) transactions
- Hence, charges can be disputed later by consumers. Because the merchant never sees the credit card, nor receives a hand-signed agreement to pay from the customer, when disputes arise, the merchant faces the risk that the transaction may be disallowed and reversed, even though he has already shipped the goods or the user has downloaded a digital product.

FIGURE 5.14 **HOW AN ONLINE CREDIT CARD TRANSACTION WORKS**



- Figure 5.14 illustrates the online credit card purchasing cycle.
- There are five parties involved in an online credit card purchase:
  - consumer
  - Merchant
  - Clearinghouse
  - merchant bank (sometimes called the “acquiring bank”)
  - consumer’s card-issuing bank.

- In order to accept payments by credit card, online merchants must have a merchant account established with a bank or financial institution.

Process:

- an online credit card transaction begins with a purchase

(1). When a consumer wants to make a purchase—

  - He or she adds the item to the merchant’s shopping cart.

(2). a secure tunnel through the Internet is created using SSL/TLS.

  - Using encryption, SSL/TLS secures the session during which credit card information will be sent to the merchant and protects the information from interlopers on the Internet.
  - SSL does not authenticate either the merchant or the consumer. The transacting parties have to trust one another.

(3). Once the consumer credit card information is received by the merchant—

  - the merchant software contacts a clearinghouse.
  - a clearinghouse is a financial intermediary that authenticates credit cards and verifies account balances.

(4). The clearinghouse contacts the issuing bank to verify the account information .

(5). Once verified, the issuing bank credits the account of the merchant at the merchant’s bank (usually this occurs at night in a batch process)

(6). The debit to the consumer account is transmitted to the consumer in a monthly statement

Limitations of Online Credit Card Payment Systems

- The most important limitations involve security, merchant risk, administrative and trans-

action costs, and social equity.

- **poor security**. Neither the merchant nor the consumer can be fully authenticated. The merchant could be a criminal organization designed to collect credit card numbers, and the consumer could be a thief using stolen or fraudulent cards. The risk facing merchants is high: consumers can repudiate charges even though the goods have been shipped or the product downloaded. The banking industry attempted to develop a secure electronic transaction (SET) protocol, but this effort failed because it was too complex for consumers and merchants.
- The administrative costs of setting up an online credit card system and becoming authorized to accept credit cards are high. Transaction costs for merchants also are significant—roughly 3% of the purchase plus a transaction fee of 20–35 cents per transaction, plus other setup fees
- Credit cards are not very democratic, even though they seem ubiquitous. Millions of people do not have credit cards. Many cannot afford cards or who are considered poor risks because of low incomes.
- In countries like Nepal, where consumers are not adopted to online transactions and not very professional to pay in timely manner, banks are reluctant to issue

ALTERNATIVE ONLINE PAYMENT SYSTEMS

- The limitations of the online credit card system have opened the way for the development of a number of alternative online payment systems.

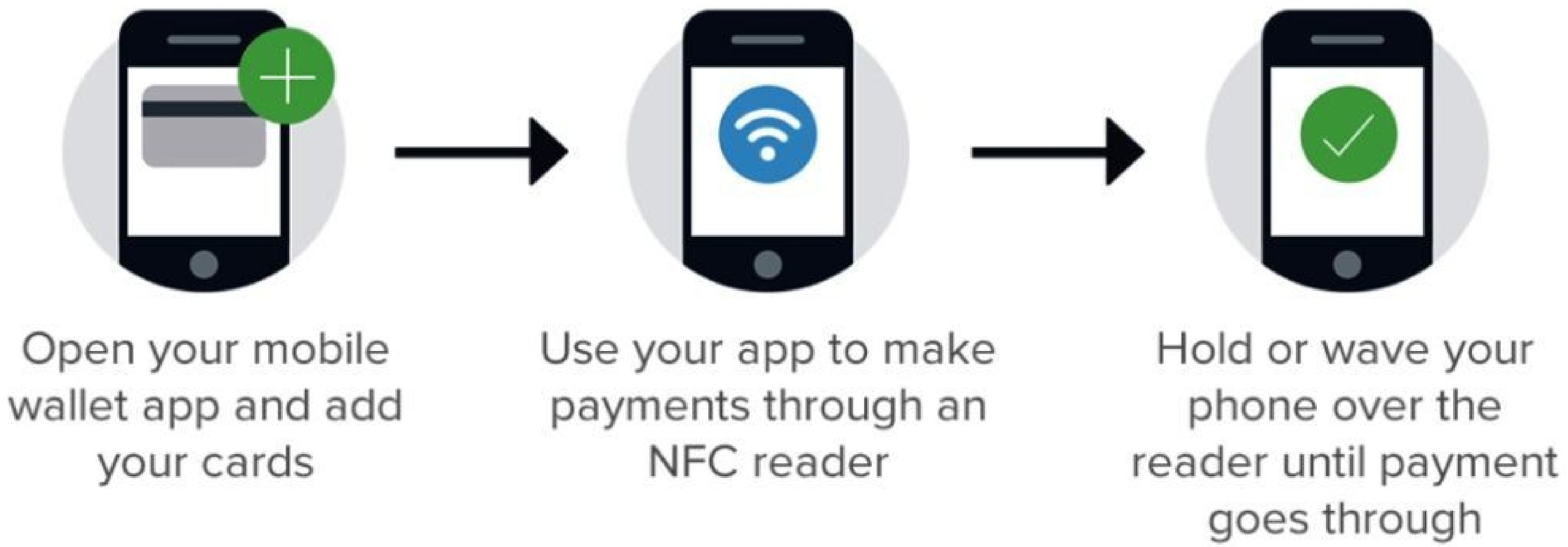
1. **online stored value payment system**
2. **Smartphone wallet**
3. **Social/mobile peer-to-peer payment systems**
4. **Digital Cash And Virtual Currencies**

1. online stored value payment system

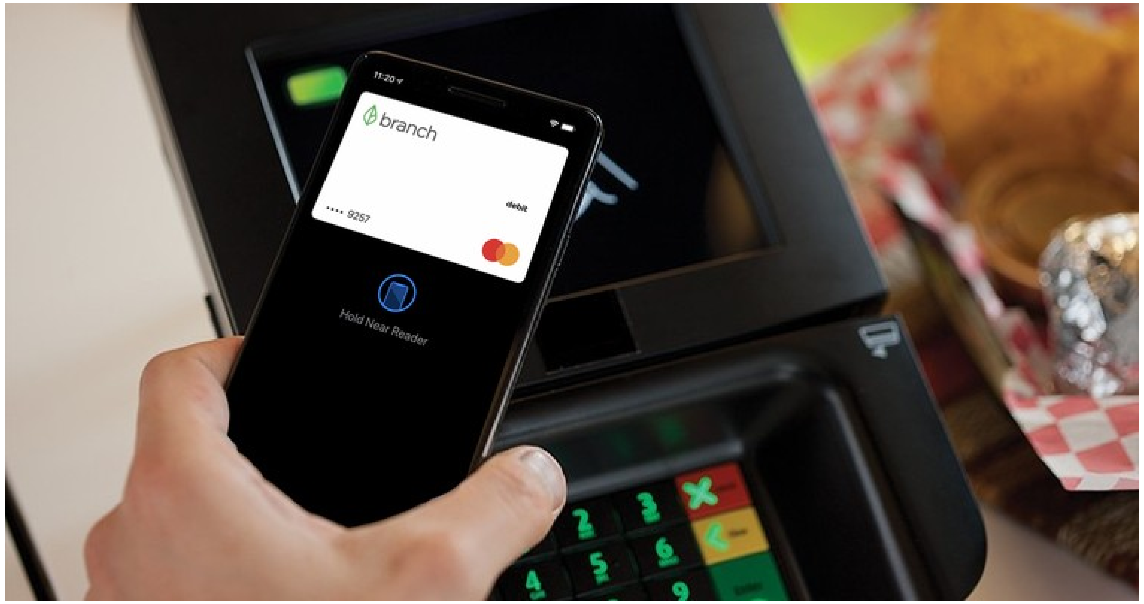
- permits consumers to make instant, online payments to merchants and other individuals based on value stored in an online account using their bank account or credit/debit cards.
- Example : PayPal enables individuals and businesses with e-mail accounts to make and receive payments up to a specified limit.
- In Nepal ??

2. Smartphone wallet

- The use of mobile devices as payment mechanisms is already well established .
- Near field communication (NFC) is the primary enabling technology for mobile payment systems.
- Near field communication (NFC) is a set of short-range wireless technologies used to share information among devices within about 2 inches of each other (50 mm).
- NFC devices are either powered or passive. A connection requires one powere one target device, o requests from the init



- NFC peer-to-peer communication is possible where both devices are powered. Consumers can swipe their NFC-equipped phone near a merchant’s reader to pay for purchases.
- In September 2014, Apple introduced the iPhone 6, which is equipped with NFC chips designed to work with Apple’s mobile payments platform, Apple Pay. Apple Pay is able to be used for mobile payments at the point-of-sale at a physical store as well as online purchases using an iPhone.
- Other competitors in NFC-enabled mobile payments include Android Pay, Samsung Pay, PayPal, etc.

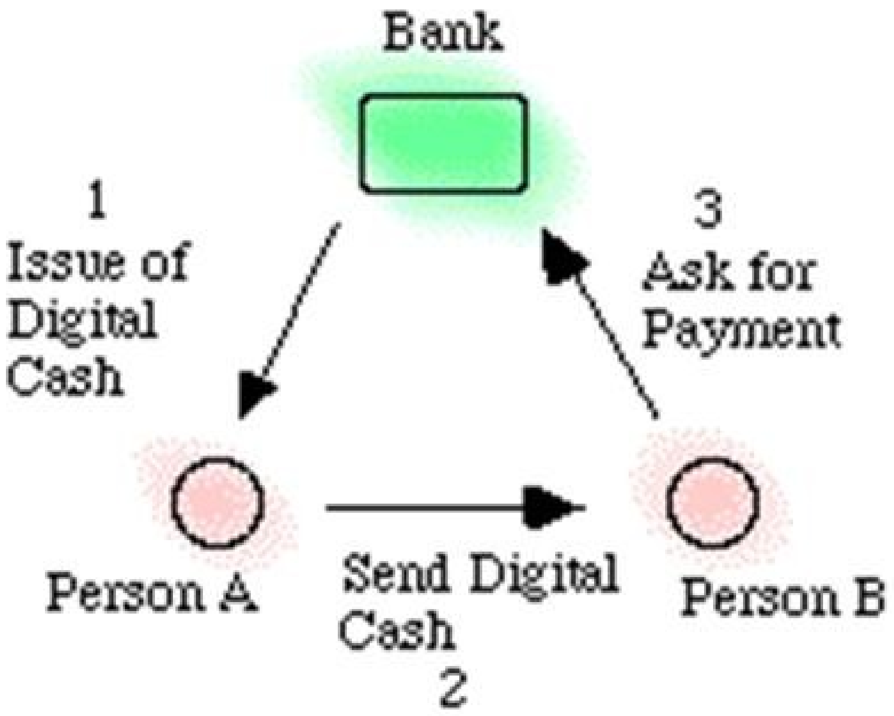


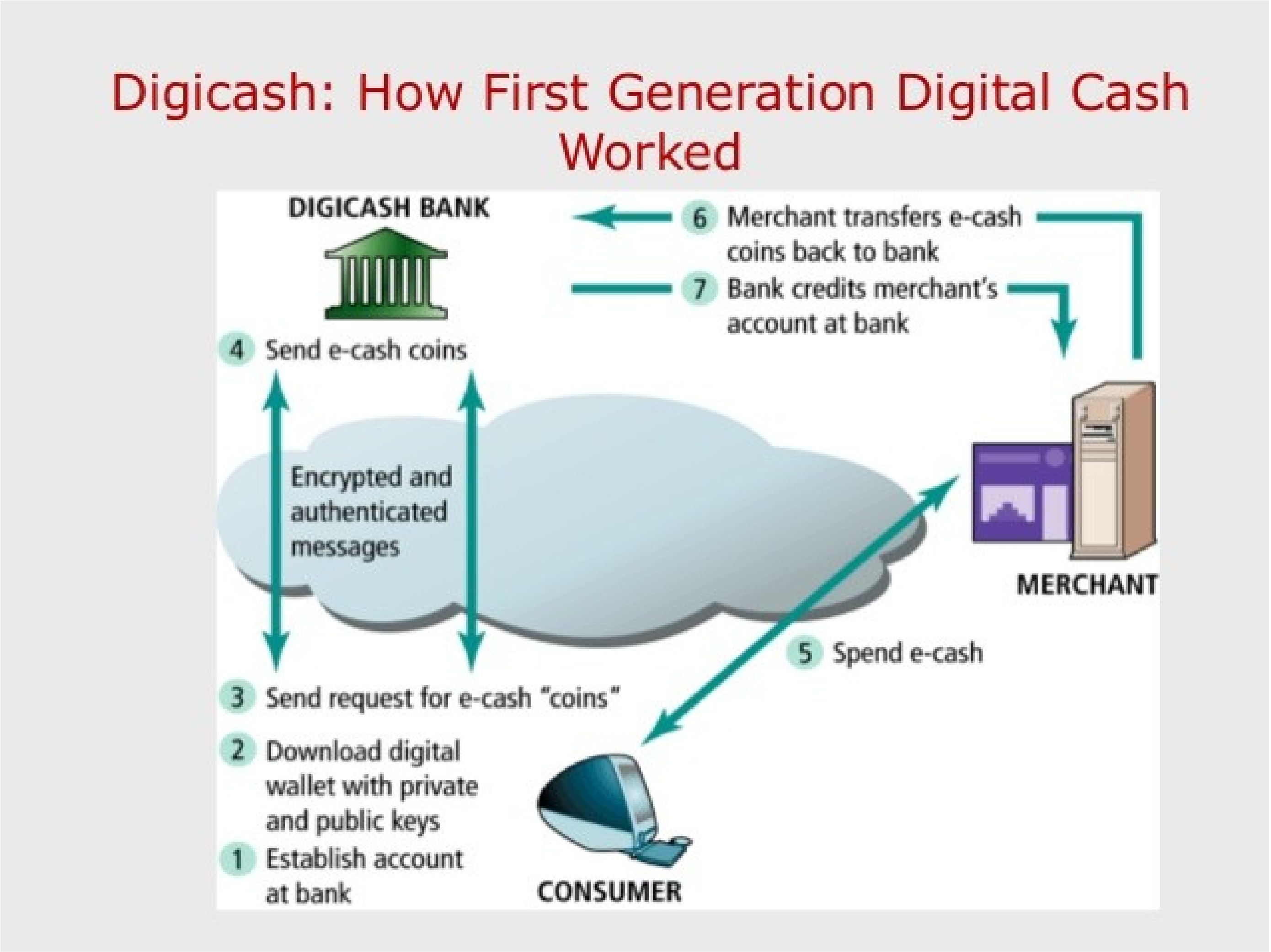
3. Social/mobile peer-to-peer payment systems

- Enable users to send another person money through a mobile application or website, funded by a bank debit card. (Rather than payment at physical point-of-sale)
- Examples: Venmo, Square Cash, Snapcash, Google Wallet, and the Facebook Messenger Payment service.
- There is no charge for this service. Venmo, owned by PayPal, is particularly popular, with its success in part due to its integration with Facebook and its social network newsfeed, which lets users see when friends are paying other friends or paying for products and services.
- In 2016, Facebook and PayPal announced that Facebook subscribers could use PayPal to purchase goods and services, with notifications coming through Facebook Messenger.

4. E-cash/Digital Cash

Traditional e-cash :





Cryptocurrency

- Cryptocurrency Digital cash is based on an algorithm that generates unique authenticated tokens representing cash value that can be used “in the real world.”
- Cryptocurrency is anonymous and there is no central controlling body (e.g. bank). The community of users themselves control the currency.
- The currency is protected by cryptographic security and transactions are maintained in a distributed ledger (called blockchain). Each peer keeps the ledger and any transaction is verified by users themselves.
- Bitcoin is the best known example of such digital cash.
- Bitcoins are encrypted numbers (sometimes referred to as cryptocurrency) that are generated by a complex algorithm using a peer-to-peer network in a process referred to as “mining” that requires extensive computing power.
- Like real currency, Bitcoins have a fluctuating value tied to open-market trading. Like cash, Bitcoins are anonymous—they are exchanged via a 34-character alphanumeric address that the user has, and do not require any other identifying information.
- Bitcoins have recently attracted a lot of attention as a potential money laundering tool for cybercriminals and illicit drug markets like Silk Road, and have also been plagued by security issues, with some high-profile heists.
- Nonetheless, there are companies now using Bitcoins as a legitimate alternative payment system.

Electronic Billing Presentment and Payment (EBPP) System

- Electronic billing presentment and payment (EBPP) systems are systems that enable the online delivery and payment of monthly bills.
- EBPP services allow consumers to view bills electronically using either their desktop PC or mobile device and pay them through electronic funds transfers from bank or credit card accounts.
- More and more companies are choosing to issue statements and bills electronically, rather than mailing out paper versions, especially for recurring bills such as utilities, insurance, and subscriptions.
- Online billing has huge market opportunity/potential. Mobile bill payments are surging.
- One major reason for the surge in EBPP usage is that companies are starting to realize how much money they can save through online billing.
- Companies are discovering that a bill is both a sales opportunity and a customer retention opportunity, and that the electronic medium provides many more options when it comes to marketing and promotion.
- Example: In Nepal, we can pay electricity bill online. We can view bill, pay online and even get an electronic receipt (via e-sewa mobile wallet)

EBPP BUSINESS MODELS

There are four EBPP business models:

1. online banking
2. biller-direct
3. Mobile
4. consolidator.

5. The **online banking model** is the widely used today.
   - Consumers share their banking or credit card credentials with the merchant and authorize the merchant to charge the consumer's bank account.

- How it works?
  - ➢ User/customer creates account/profile with the merchant website.
  - ➢ Customer sets credit card (or bank account number) in the profile.
  - ➢ Customer grants permission to the merchant to deduct money from customer account.
  - ➢ Whenever there is bill to pay, the merchant’s system automatically performs payment transaction and money is deducted from customer’s bank account.

- This model has the advantage of convenience for the consumer because the payments are deducted automatically, usually with a notice from the bank or the merchant that their account has been debited.

2. In the **biller-direct model**, consumers are sent bills by e-mail notification, and go to the merchant’s website to make payments using their banking credentials.

- This model has the advantage of allowing the merchant to engage with the consumer by sending coupons or rewards.
- The biller-direct model is a two-step process, and less convenient for consumers.

3. The **mobile model** allows consumers to make payments using mobile apps, once again relying on their bank credentials as the source of funds.

- Consumers are notified of a bill by text message and authorize the payment.
- An extension of this is the social-mobile model, where social networks like Facebook integrate payment into their messaging services.
- The mobile model has several advantages,
    - the convenience for consumers of paying bills while using their phones
    - speed with which bills can be paid in a single step.
    - This is the fastest growing form of EBPP. In 2016, Facebook and PayPal announced a deal in which Facebook users can pay for purchases on Facebook using PayPal

4. In the **consolidator model**, a third party, such as a financial institution or a focused portal such as Intuit’s Paytrust, Fiserv’s MyCheckFree, Mint Bills, and others, aggregates all bills for consumers and permits one-stop bill payment.

- This model has the advantage of allowing consumers to see all their bills at one website or app.
- However, because bills come due at different times, consumers need to check their portals often.
- The consolidator model faces several challenges.
    - For billers, using the consolidator model means an increased time lag between billing and payment,
    - and also inserts an intermediary between the company and its customer.

Online auctions

- Auctions are widely used in the e-commerce

1. consumer-to-consumer (C2C) auctions
   - The most widely known auctions are c2c auctions
   - The auction house is simply an intermediary market maker, providing a forum where consumers—buyers and sellers—can discover prices and trade.
   - The market leader in C2C auctions is eBay.

2. business-to-consumer (B2C) auctions
   - Less well known auctions are business-to-consumer (B2C) auctions, where a business owns or controls assets and uses dynamic pricing to establish the price.
   - Auctions constitute a significant part of B2B e-commerce recently
   - Auctions are not limited to goods and services. They can also be used to allocate resources, and bundles of
   - resources, among any group of bidders.
   - For instance, if we wanted to establish an optimal schedule for assigned tasks in an office among a group of clerical workers, an auction in which workers bid for assignments would come close to producing a nearly optimal solution in a short amount of time.
   - Auctions are ways of allocating resources among independent agents (bidders).

BENEFITS OF AUCTIONS

- **Liquidity**: Sellers can find willing buyers, and buyers can find sellers. Sellers and buyers can be located anywhere around the globe. Buyers and sellers can find a global market for rare items that would not have existed before the Internet
- **Price discovery**: Buyers and sellers can quickly and efficiently develop prices for items that are difficult to assess, where the price depends on demand and supply, and where the product is rare.
- Price transparency: Public Internet auctions allow everyone in the world to see the asking and bidding prices for items.
- **Market efficiency**: Auctions can, lead to reduced prices, and hence reduced profits for merchants, leading to an increase in consumer welfare.
- **Lower transaction costs**: Online auctions can lower the cost of selling and purchasing products, benefiting both merchants and consumers. Like other Internet markets, such as retail markets, Internet auctions have very low transaction costs.
- **Consumer aggregation**: Sellers benefit from large auction sites' ability to aggregate a large number of consumers who are motivated to purchase something in one marketspace.
- **Network effects**: The larger an auction site becomes in terms of visitors and products for sale, the more valuable it becomes as a marketplace for everyone by providing

Risks and Costs of Auctions

- There are a number of risks and costs involved in participating in auctions.
- In some cases, auction markets can fail—like all markets at times.
- Some of the more important risks and costs are:
  - **Delayed consumption costs**: Internet auctions can go on for days, and shipping will take additional time.
  - **Monitoring costs**: Participation in auctions requires your time to monitor bidding.
  - **Equipment costs**: Internet auctions require you to purchase a computer system and pay for Internet access.
  - **Trust risks**: Online auctions are a significant source of Internet fraud. Using auctions increases the risk of experiencing a loss.
  - **Fulfillment costs**: Typically, the buyer pays fulfillment costs of packing, shipping, and insurance, whereas at a physical store these costs are included in the retail price.

How to reduce costs and risks ?

- To reduce consumer participation costs and trust risk, auction sites attempt to solve the trust problem by providing a rating system in which previous customers rate sellers based on their overall experience with the merchant.
- A partial solution to high monitoring costs is, ironically, fixed pricing.
- At eBay, consumers can reduce the cost of monitoring and waiting for auctions to end by simply clicking on the Buy It Now button and paying a premium price.
- The difference between the Buy It Now price and the auction price is the cost of monitoring.

TYPES AND EXAMPLES OF AUCTIONS

1. **English auction**
    - most common form of auction on eBay.
    - Typically, there is a single item up for sale from a single seller. There is a time limit when the auction ends, a reserve price below which the seller will not sell (usually secret), and a minimum incremental bid set.
    - Multiple buyers bid against one another until the auction time limit is reached.
    - The highest bidder wins the item (if the reserve price of the seller has been met or exceeded).
    - English auctions are considered to be seller-biased because multiple buyers compete against one another (usually anonymously).
2. **Dutch Internet auction**
    - This auction format is perfect for sellers that have many identical items to sell.
    - Sellers start by listing a minimum price, or a starting bid for one item, and the number of items for sale.
    - Bidders specify both a bid price and the quantity they want to buy.
    - The uniform price reigns.
    - Winning bidders pay the same price per item, which is the lowest successful bid.
    - This market clearing price can be less than some bids. If there are more buyers than items, the earliest successful bids get the goods.

- In general, high bidders get the quantity they want at the lowest successful price, whereas low successful bidders might not get the quantity they want (but they will get something).

3. **Name Your Own Price auction**

- was pioneered by Priceline, and is the second most popular auction format on the Web.
- Although Priceline also acts as an intermediary, buying blocks of airline tickets, hotel rooms, and vacation packages at a discount and selling them at a reduced retail price or matching its inventory to bidders, it is best known for its Name Your Own Price auctions
- In “Name your own price” auction, users specify what they are willing to pay for goods or services, and multiple providers bid for their business.
- Prices do not descend and are fixed: the initial consumer offer is a commitment to purchase at that price.

4. Penny auction

- also known as a bidding fee auction
- To participate in a penny auction , one typically must pay (small amount of money) the penny auction site for bids ahead of time.
- Once we have purchased the bids, we can use them to bid on items listed by the penny auction site (unlike traditional auctions, items are owned by the site, not third parties).
- Items typically start at or near zero (0) and each bid raises the price by a fixed amount, usually just a penny.
- Auctions are timed, and when the time runs out, the last and highest bidder wins the item.
- Unlike a traditional auction, it costs money to bid, and that money is gone even if the bidder does not win the auction.
- The bidder’s cumulative cost of bidding must be added to the final price of a successful bid to determine the true cost of the item.
- Bidders may find that they spend far more than they intended.
- Examples of penny auction sites include QuiBids, Beezid, and HappyBidDay.

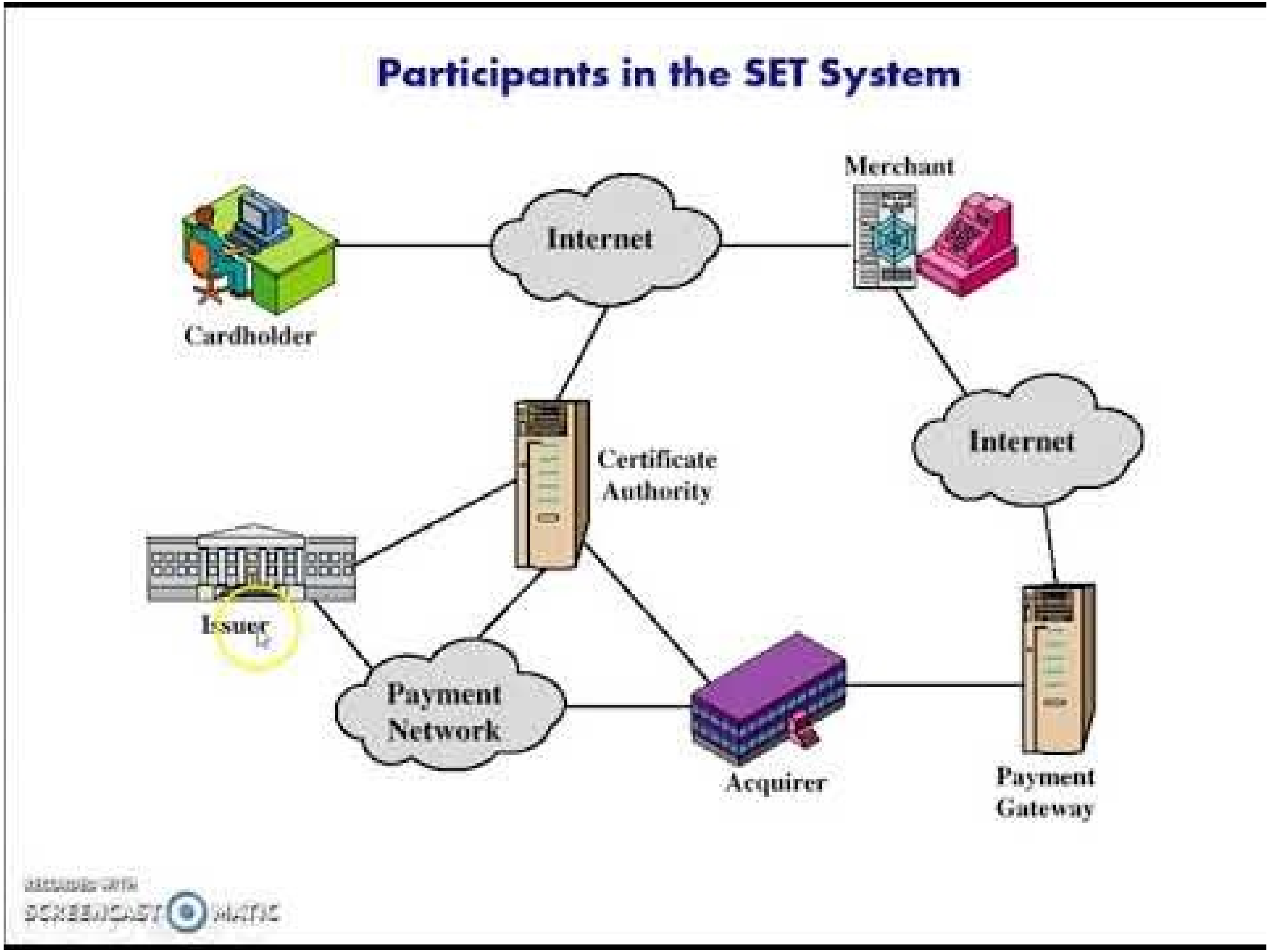
TABLE 11.6 FACTORS TO CONSIDER WHEN CHOOSING AUCTIONS

CONSIDERATIONS	DESCRIPTION
Type of product	Rare, unique, commodity, perishable
Stage of product life cycle	Early, mature, late
Channel-management issues	Conflict with retail distributors; differentiation
Type of auction	Seller vs. buyer bias
Initial pricing	Low vs. high
Bid increment amounts	Low vs. high
Auction length	Short vs. long
Number of items	Single vs. multiple
Price-allocation rule	Uniform vs. discriminatory
Information sharing	Closed vs. open bidding

Secure Electronic Transaction (SET)

- Developed by MastarCard and Visa in 1996 as a method to secure payment card transactions over open networks.
- Secure Electronic Transaction is an open-source encryption and security specification which is designed for protecting credit card transaction on the internet.
- SET is not a payment system. It is a set of security protocols and format that ensures that using online payment transaction on the internet is secure.
- SET provides a secure environment for all the parties that are involved in the e-commerce transaction.
- It also ensures confidentiality.
- It provides authentication (of both merchant and custormer) through digital certificates.
- SET protocol restricts revealing of credit card details to merchants thus keeping hackers and thieves at bay
- Provide Message Confidentiality : Confidentiality refers to preventing unintended people from reading the message being transferred. SET implements confidentiality by using encryption techniques.
- Provide Message Integrity : SET doesn’t allow message modification with the help of signatures. Messages are protected against unauthorized modification using RSA digital signatures.

SET participants



- **Cardholder**: A cardholder is an authorized holder of the payment card. The card can be a Master card or Visa which has been issued by an issuer.
- **Merchant**: A merchant is any person or organization who wants to sell their goods and services to cardholders. Note that a merchant must have a relationship with the acquirer to accept the payment through the internet.
- **Issuer**: An issuer is a financial organization such as a bank who issues payment card – Master card or visa to user or cardholder. The issuer is responsible for the cardholder’s debt payment.
- **Acquirer**: This is a financial organization that has a relationship with the merchant for processing the card payment authorization and all the payments. An acquirer is part of this process because the merchant can accept credit cards of more than one brand. It also provides an electronic fund transfer to the merchant account.
- **Payment Gateway**: For payment authorization, payment gateway acts as an interface between secure electronic transactions and existing card payment networks. The merchant exchange the Secure Electronic Transaction message with payment gateway through the internet. In response to that, the payment gateway connects to the acquirer’s system by using a dedicated network line.
- **Certification Authority**: It is a trusted authority that provides public-key certificates to the cardholders, payment gateways, and merchants.

How SET Works?

Step 1: Customer Open an Account

The customer opens a credit card account like a master card or visa with a bank (i.e. issuer) that supports electronic payment transactions and the secure electronic transaction protocol.

Step 2: Customer Receive a Certificate

Once the customer identity is verified (Verification can be done by using a passport, business documents or other documents), it receives a digital certificate which is issued by CA (Certificate Authority). This certificate contains customer details like name, public key, expiry date, certificate number, etc.

Step 3: Merchant Receives a Certificate

The merchant who wants to accept certain brands of a credit card must process a digital certificate for trustworthiness.

Step 4: Customer Place an Order

It is a shopping cart process, where customers borrowed an item from the available list, can search the specific item according to requirements, and place the order. Once the customer places the orders, the merchant in return sends the details of the order such as a list of items selected, their quantity and price, total bill, etc to maintain a record of order at the customer site.

Step 5: Merchant is Verified

Merchant also sends a digital certificate to the customer to ensues the customers that he or she is dealing with an authorized or valid merchant.

Step 6: The Order and Payment Details Are Sent

Along with the customer’s digital certificate customer also sends an order and payment details to the merchant. The order part is used to confirm the transaction with the reference of items that are mentioned in the order form. The payment part contains the credit card(master card or visa) details. This payment information is in encrypted form, even merchant cannot read it. The customer certificate ensures the merchant of a customer’s identity.

Step 7: Merchant Requests Payment Authorization

Once the merchant gets the payment details from the customer, it transfers it to the payment gateway via the acquirer and requests the payment gateway to authorize the payment details. This process ensures start the customer credit card is valid and the credit limit is not breached.

Step 8: Payment Gateway Authorizes the Payment

Using the credit card information which is received from the merchant, payment gateway cross verify the credit card of the customer with the help of the issuer. Based on the verification result, it either authorizes the payment or rejects the payment.

Step 9: Merchant Confirm the Order

Assuming that the payment gateway authorizes the payment, merchants send confirmation of order to the customer.

Step 10: Merchant Provides a Goods and Services

Now the merchant provides goods and services according to the customer's order.

Step 11: Merchant Request Payment

The merchant sends a request to the payment gateway for making payment. After that, payment gateway interacts with various financial organizations such as issuer, acquirer and the clearinghouse to effect the payment from the customer's account to the merchant's account.

Self Study

- **Status of e-payment systems in Nepal**